

Trade Intensity Index: Degree to which 2 countries trade more intensively with each other than with other trading partners.
 T_{ij} = share of country i's export going to country j relative to share of j's import from the world.

$$T_{ij} = \frac{x_{ij} / x_i}{M_j / T}$$

x_{ij} = Country i's export to country j.
 x_i = total exports in country i
 M_j = total imports in country j
 T = total world trade

T_{ij} = country i's export going to country j relative to country j's total imports from the world

$\Rightarrow$ percent of country going to j's export

country i's total exports / total world trade

Trade Complimentary Index: The extend to which the export pattern of one country matches the commodity mixture of import pattern of its trading partner.
 C_{ij} = The extend to which the export specialization in one country matches the import specialization in another country.

$$C_{ij} = \sum_k \left(\frac{x_{ik}}{x_i} \right) \left(\frac{T}{T_k} \right) \left(\frac{M_{jk}}{M_j} \right)$$

k = individual commodities
 x_{ik} = country i's export of k commodity
 x_i = country i's total export
 T = total world trade
 T_k = total world trade for product k
 M_{jk} = country j's import of product k
 M_j = country j's total imports

Trade Biased Index: Serve as proxy of the relative factors which influence Various trade flows on a bilateral basis

Lingling Yang

a = country a
b = country b
k = product k
 w_k = world export of k
W = total world export

Trade complimentary index of single product k between country a & b

weight of a given export. $\frac{w_k}{W}$

Export $RCA_{xa}^k = \left(\frac{x_a^k}{x_a} \right) / \left(\frac{w_k}{W} \right)$

Import $RCA_{mb}^k = \left(\frac{m_b^k}{m_b} \right) / \left(\frac{w_k}{W} \right)$

$C_{ab}^k = RCA_{xa}^k \cdot RCA_{mb}^k$

Weighted avg. of all product.

$$C_b^a = \sum_k C_{ab}^k \frac{w_k}{W}$$

$$= \sum_k RCA_{xa}^k \cdot RCA_{mb}^k \left(\frac{w_k}{W} \right)$$

$$= \sum_k \left(\left(\frac{x_a^k}{x_a} \right) / \left(\frac{w_k}{W} \right) \right) \cdot \left(\left(\frac{m_b^k}{m_b} \right) / \left(\frac{w_k}{W} \right) \right) \cdot \left(\frac{w_k}{W} \right)$$

Tan Fen

$w_{mk} = w_{xk}$

$$RCA_{xik} = \left(\frac{x_{ik}}{x_i} \right) / \left(\frac{w_{xk}}{W_x} \right)$$

$$RCA_{mjk} = \left(\frac{m_{jk}}{m_j} \right) / \left(\frac{w_{mk}}{W_m} \right)$$

$$C_k = RCA_{xik} \cdot RCA_{mjk}$$

$$C_k = \left(\left(\frac{x_{ik}}{x_i} \right) / \left(\frac{w_{xk}}{W_x} \right) \right) \cdot \left(\left(\frac{m_{jk}}{m_j} \right) / \left(\frac{w_{mk}}{W_m} \right) \right)$$

$$C_{ij} = \sum_k \left(\frac{x_{ik}}{x_i} \right) \left(\frac{T}{T_k} \right) \left(\frac{m_{jk}}{m_j} \right)$$

$$\sum RCA_{xa}^k RCA_{mb}^k \frac{w_k}{W}$$

$$= \sum \left(\left(\frac{x_a^k}{x_a} \right) / \left(\frac{w_k}{W} \right) \right) \cdot \left(\left(\frac{m_b^k}{m_b} \right) / \left(\frac{w_k}{W} \right) \right) \cdot \left(\frac{w_k}{W} \right)$$

$$\Rightarrow \sum \frac{x_a^k}{x_a} \cdot \frac{m_b^k}{m_b} \div \frac{w_k}{W} \Leftarrow \text{Flip this}$$

$$= \sum \frac{x_a^k}{x_a} \cdot \frac{m_b^k}{m_b} \cdot \frac{W}{w_k}$$

$$= \sum \frac{x_a^k}{x_a} \cdot \frac{W}{w_k} \cdot \frac{m_b^k}{m_b} \Rightarrow \left(\frac{x_{ik}}{x_i} \right) \left(\frac{T}{T_k} \right) \left(\frac{m_{jk}}{m_j} \right)$$